SINGAPORE SCHOOL

CONTINUAL ASSESSMENT 1

Secondary 1

Computer Science (Part 2) — ANSWER KEY

2026-27

Date: ___ / ___ / ___ **Name: _____**

Instructions for students:

- **Start answering the questions on time.**
- **Follow all the instructions carefully.**
- **Answer all questions.**
- **Revise**

CA 1 Marks (Part – 2)	**25**
CW	**05**
Total Marks Obtained	**30**

1. Choose the correct answer for each of the following multiple-choice questions. (1 x 10 = 10)

1. How many bits make up 1 Byte of data?

A) 4 bits
B) 8 bits
C) 16 bits
D) 1,024 bits

➔ Answer: B) 8 bits

2. How many Hertz (Hz) are there in 1 Kilohertz (1 kHz)?

A) 100 Hz
B) 1,000 Hz
C) 1,024 Hz
D) 1,000,000 Hz

➔ Answer: B) 1,000 Hz

3. How many Bytes are there in 1 KiB (Kibibyte)?

A) 1,000 Bytes
B) 1,024 Bytes
C) 8,192 Bytes
D) 1,048,576 Bytes

➔ Answer: B) 1,024 Bytes

4. In standard 8-bit ASCII encoding, how many Bytes are needed to store 1 character?

A) 1 Byte
B) 2 Bytes
C) 4 Bytes
D) 8 Bytes

➔ Answer: A) 1 Byte (8 bits = 1 Byte)

5. In 16-bit Unicode encoding, how many Bytes are required to store a single character?

A) 1 Byte
B) 2 Bytes
C) 8 Bytes
D) 16 Bytes

➔ Answer: B) 2 Bytes (16 bits ÷ 8 = 2 Bytes)

6. If a bitmap image uses a colour depth of 4 bits per pixel, what is the maximum number of colours it can display?

A) 4 colours
B) 8 colours
C) 16 colours
D) 256 colours

➔ Answer: C) 16 colours (2^4 = 16 colours)

7. How many bits per pixel (colour depth) are needed for an image with 256 colours?

A) 2 bits
B) 4 bits
C) 8 bits (1 Byte)
D) 24 bits

➔ Answer: C) 8 bits (1 Byte) (2^8 = 256)

8. Which formula correctly calculates the uncompressed file size of a bitmap image in bits?

A) Width + Height + Colour Depth
B) Width × Height × Colour Depth
C) (Width × Height) ÷ Colour Depth
D) Sample Rate × Width × Height

➔ Answer: B) Width × Height × Colour Depth

9. When calculating the file size of a STEREO digital audio recording, what multiplier is used for the number of channels?

A) 1 B) 2
C) 4 D) 8

➔ Answer: B) 2 (Mono = 1 channel, Stereo = 2 channels)

10. A sample rate of 44.1 kHz is equal to how many samples per second?

A) 441 Hz B) 4,410 Hz
C) 44,100 Hz D) 441,000 Hz

➔ Answer: C) 44,100 Hz (44.1 × 1,000 = 44,100 Hz)

2. Convert each of the following frequency and data storage measurements. (Picked from Worksheet Q1, Q2 & Q3) **(1 x 3 = 3)**

(a) Convert 8 kHz into Hz, and 2.5 MHz into kHz.

Worked Solution:
8 kHz = 8 × 1,000 = 8,000 Hz | 2.5 MHz = 2.5 × 1,000 = 2,500 kHz

(b) Convert 64 bits into Bytes, and 400 Bytes into bits.

Worked Solution:
64 bits ÷ 8 = 8 Bytes | 400 Bytes × 8 = 3,200 bits

(c) Convert 5 KiB into Bytes (using 1,024), and 2,048 KiB into MiB.

Worked Solution:
5 KiB × 1,024 = 5,120 Bytes | 2,048 KiB ÷ 1,024 = 2 MiB

3. Solve the following TEXT file size problems. Show your working in the box. **(1.5 x 2 = 3)** (Picked from Worksheet Q5 & Q6)

(a) A student types the phrase "HELLO SECONDARY 1" using 8-bit ASCII encoding. Count the total characters (including spaces) and calculate the file size in bits and Bytes.

Worked Solution:
Characters = 17 (including 2 spaces). Size in bits = 17 × 8 = 136 bits. Size in Bytes = 17 × 1 = 17 Bytes.

(b) A plain text essay contains 4,096 characters and is saved using 8-bit ASCII encoding. Calculate the file size in Bytes and in KiB (dividing by 1,024).

Worked Solution:
Size in Bytes = 4,096 × 1 = 4,096 Bytes. Size in KiB = 4,096 ÷ 1,024 = 4 KiB.

4. Solve the following BITMAP IMAGE file size problems. Show your working in the box. (Picked from Worksheet Q9 & Q11) **(1.5 x 2 = 3)**

(a) A small game icon has a resolution of 100 × 80 pixels and uses a colour depth of 8 bits per pixel. Calculate the image file size in bits, Bytes, and KB (dividing by 1,000).

Worked Solution:
Bits = 100 × 80 × 8 = 64,000 bits. Bytes = 64,000 ÷ 8 = 8,000 Bytes. KB = 8,000 ÷ 1,000 = 8 KB.

(b) A retro graphic has a resolution of 512 × 256 pixels and displays 16 colours (4 bits per pixel). Calculate the exact file size in Bytes and KiB (dividing by 1,024).

Worked Solution:
Bits = 512 × 256 × 4 = 524,288 bits. Bytes = 524,288 ÷ 8 = 65,536 Bytes. KiB = 65,536 ÷ 1,024 = 64 KiB.

5. Solve the following DIGITAL SOUND file size problems. Show your working in the box. (Picked from Worksheet Q15 & Q16) **(1.5 x 2 = 3)**

(a) A 10-second mono voice note is recorded with a sample rate of 8,000 Hz and a sample resolution of 8 bits. Calculate the file size in bits, Bytes, and KB (dividing by 1,000).

Worked Solution:
Bits = 8,000 × 8 × 10 × 1 (mono) = 640,000 bits. Bytes = 640,000 ÷ 8 = 80,000 Bytes. KB = 80,000 ÷ 1,000 = 80 KB.

(b) A 30-second stereo sound clip is recorded at a sample rate of 20 kHz (20,000 Hz) with a sample resolution of 16 bits (2 Bytes). Calculate the file size in Bytes and MB (dividing by 1,000,000).

Worked Solution:
Bytes = 20,000 × 2 Bytes × 30 seconds × 2 (stereo) = 2,400,000 Bytes. MB = 2,400,000 ÷ 1,000,000 = 2.4 MB.

6. Multimedia Challenge Problem. Show all steps clearly in the box. (Picked from Worksheet Q20) **(1 x 3 = 3)**

A student creates a multimedia slide containing three elements:

• Text: 2,000 characters encoded in 8-bit ASCII (1 Byte per character).

• Image: One bitmap picture of 500 × 400 pixels with 8-bit colour depth (1 Byte per pixel).

• Audio: One 5-second mono voice clip sampled at 10 kHz (10,000 Hz) with 8-bit resolution (1 Byte per sample).

Calculate the size of each element in KB and find the total combined file size in KB (dividing by 1,000).

Worked Solution:
1. Text = 2,000 Bytes = 2 KB
2. Image = 500 × 400 × 1 = 200,000 Bytes = 200 KB
3. Audio = 10,000 × 1 × 5 × 1 = 50,000 Bytes = 50 KB
➔ Total Combined File Size = 2 + 200 + 50 = 252 KB (252,000 Bytes)